MILITARY SPECIFICATION

PROJECTILE, GUIDED, 155MM, XM712
ROLL RATE SENSOR FOR

This specification is approved for use by U.S. Army Armament Research and Development Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification establishes the requirements for the manufacture and acceptance of the Roll Rate Sensor (RRS), critical item.

2. APPLICABLE DOCUMENTS

2.1 Government documents.- Except as noted, the following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-P-63223	-	Projectile, Guided, 155mm, XM712
MIL-P-63224	-	Projectile, Guided, 155mm, XM712, Guidance Section for
MIL-P-63227	-	Projectile, Guided, 155mm, XM712, Packaging for

STANDARDS

Military

MIL-STD-105	-	Sampling Procedures and Tables For Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-810	-	Environmental Test Methods

DRAWINGS

Army

9303350	-	Roll Rate Sensor Assembly
9303362	-	Cover, Rate Sensor Assembly

FSC: 1320

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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9303364	-	Processor Assembly
9303366	-	Board Assembly, Bridge Amplifier
9303368	-	Board Assembly, Pump Oscillator
9303370	-	Acceptance Test, Pump Oscillator Assembly
9303371	-	Acceptance Test, Bridge Amplifier Assembly
9303372	-	Acceptance Test for Electronic Interconnect Assembly
9303380	-	Block, Mounting
9303382	-	Housing, Rate Sensor
9303390	-	Roll Rate Sensor Installation

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specified procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Item definition.- The Roll Rate Sensor (RRS) is an angular rate sensor with signal processing circuitry, used to sense projectile roll rate and generate a command signal for control of the projectile roll attitude. Contained in the RRS is a continuous, laminar flow gas jet, which impinges on two hot wire resistance sensors. The gas is pumped in a closed cycle by an electrically excited, vibrating piezoelectric diaphragm. Angular rate inputs cause the gas jet to deflect laterally, resulting in a differential cooling of the hot wires. The associated changes in resistance of the hot wires, which are connected in a bridge circuit, produce a signal proportional to the input rate.

3.2 CHARACTERISTICS.-

3.2.1 Performance.- The RRS shall comply with the requirements of this specification during the operational environment specified in 3.2.2 after being subjected to the nonoperational and launch shock environments specified in 3.2.2.

3.2.1.1 Electrical requirements.- The RRS shall provide the output signal specified herein when the input roll rates and input voltages specified herein are applied.

3.2.1.1.1 Input signals.-

3.2.1.1.1.1 Input power voltages.- Three input power voltages shall be available to the RRS. One voltage (11.5 Vdc) shall be bounded by the time varying characteristics of Figure 5, and shall be capable of supplying 300 milliamperes (ma) during warmup and operation. The other voltages are $\pm$15.0 $\pm$ 0.75 Vdc and shall be used during operation. The +15 and -15 V sources shall be capable of supplying 100 ma. During warmup, the +15 V supply shall be +2.0 $\pm$ 2.0 Vdc and the -15 V supply shall be +0.4 $\pm$ 0.4 Vdc. Figure 1 defines the turn-on timing sequence.

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3.2.1.1.1.2 Null store signal.- The null store input signal shall start at the application of warmup power to the RRS and the duration shall be variable from 2 seconds to 49 seconds. The RRS shall null store when the null store input signal is 5.0 ± 0.4 Vdc from a maximum source current of 0.4 milliamperes. The null store signal (low) during operation is +0.1 ± 0.1 Vdc. The plus and minus 15 Vdc shall be available 1.6 ± 0.2 seconds prior to end of null store. See Figure 1.

3.2.1.1.1.3 Latch signal.- The latch signal shall initially establish the roll command output at a saturation level of +9.0 to +14.75 Vdc. Upon RRS activation, the logic shall unlatch the roll command output and provide roll command output which is dependent upon RRS roll rate output. The latch logic shall be released (unlatched) by connection of the latch input line to a -15 Vdc supply.

3.2.1.1.2 Output signals.- The RRS shall provide a roll rate output signal and a roll command output signal with characteristics as defined in the following paragraphs.

3.2.1.1.2.1 Roll rate output limits.- The roll rate output signal, as a function of input roll rate, shall be within the boundaries of Figure 2, with a load impedance of 500,000 ohms minimum.

3.2.1.1.2.2 Roll command output voltage limits.- Roll command output limits, with respect to power return, and with a load impedance of >10,000 ohms, shall be 9.0 to 14.75 Vdc for clockwise projectile roll, and -9.0 to -14.75 Vdc for counterclockwise projectile roll. The total roll angle excursion required to produce minus to plus voltage limits shall be 120 to 300 degrees. (Note: Projectile roll direction is determined by looking from the rear, or mounting face of the RRS.)

3.2.1.2 Rate signal frequency response.- The amplitude and phase frequency response of the roll rate output signal shall be constrained by the boundaries shown in Figure 3 where 0 dB is measured dc gain.

3.2.1.3 Combined error due to rate of null store.- The drift rate of the roll command output signal, or the rate signal due to the static bias plus null store rate induced bias, shall be within the boundaries shown in Figure 4. These boundaries shall apply for operational times up to 70 seconds after null store goes low. Drift rate compensation shall be optimized such that the command output is minimized at 35 seconds after null store.

3.2.1.4 Insulation resistance.- The insulation resistance between the case ground lead and all input or output leads shall be a minimum of 10 megohms at test potential of 50 ± 5 Vdc.

3.2.1.5 Activation.- With input power of 3.2.1.1.1.1 applied, and with the RRS spinning at a clockwise roll rate that will produce a roll output signal >3.0 volts, the RRS shall apply the latch control to the roll command output within 0.050 seconds after the removal of the null store signal of 3.2.1.1.1.2. The RRS shall unlatch the roll command output within 0.050

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seconds after application of the latch to the roll command output. For clockwise projectile roll rates corresponding to a roll rate output of <2.0 volts, the RRS shall apply the latch control to the roll command output within 0.050 seconds after the removal of the null store signal of 3.2.1.1.1.2. The RRS shall not unlatch the roll command output signal.

3.2.2 Environmental conditions.- The RRS shall meet the requirements of 3.2.1 during and/or after exposure to the following environmental conditions as specified.

3.2.2.1 Temperature.- The RRS shall meet the requirements of 3.2.1 at ambient temperatures between -25°F to +155°F. The RRS shall be capable of withstanding exposure to temperatures from -50°F to +160°F.

3.2.2.2 Vibration.- The RRS shall meet the requirements of 3.2.1 during a random vibration environment of 7.6 g rms, 0.04 G^2/Hz power spectral density, 20 to 2000 Hz.

3.2.2.3 Relative humidity.- The RRS shall meet the requirements of 3.2.1 while being exposed to relative humidities of up to 94 $\pm$ 4 percent at +68°F to +149°F.

3.2.2.4 Transportation vibration.- The RRS assembly assembled into a guided projectile, MIL-P-63223, and packaged into a tactical container, MIL-P-63227, shall meet requirements of 3.2.1 after exposure to the transportation vibration requirements of MIL-STD-810, Method 514.2, Procedure X.

3.2.2.5 Fragility shock.- The RRS assembly assembled into a guided projectile, MIL-P-63223, and packaged into a tactical container, MIL-P-63227 shall meet the requirements of 3.2.1 after exposure in all three axes to half sine wave mechanical shock pulses having nominal durations of 10 milliseconds, with nominal peak accelerations of 200g.

3.2.2.6 Launch shock.- The RRS shall withstand, and meet the requirements of 3.2.1, following exposure to cannon launch acceleration of 9000 $\pm$ 300 g forward along the input axis of the RRS.

3.3 Design and construction.-

3.3.1 Production drawings.- This RRS shall be fabricated and assembled in accordance with the drawings, parts lists, and other documents listed on drawing 9303350.

3.3.2 Workmanship.- All parts shall be fabricated and finished in a thoroughly workmanlike manner. Except as otherwise specified, the parts shall be free of cracks, scratches, burrs, dents, gouges, surface defects or any surface discontinuities. The items shall be free from dirt, chips, grease, corrosion, and other foreign material. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. All required marking shall be neat and sharply defined.

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3.4 First article inspection.- This specification makes provisions for first article inspection. First article samples shall meet all the requirements of this specification. Requirements for submission of first article samples by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the procuring activity. The procuring activity reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection.- The following types of inspection shall be conducted on this item:

(a) First article inspection (4.3)

(b) Quality conformance inspection (4.4)

4.3 First article inspection.-

4.3.1 Submission.- The contractor, when required by contract, shall submit a first article sample of sixteen (16) RRS assemblies for examination and test by the procuring activity in accordance with Table I. The first article sample shall have been inspected in accordance with Table II. The contractor shall additonally submit 5 complete sets of parts and assemblies as listed in "Classification of Defects and Tests", Table II, sheets 1 through 6. Environmental tests shall be conducted in the sequence indicated. Performance of each RRS assembly shall be as indicated in each test method. The testing shall be considered destructive and the tested assemblies shall not be returned to a lot for delivery. All assemblies and components shall have been produced by the contractor or furnished by a supplier and shall have been manufactured using the same production processes, procedures, and equipment which will be used in fulfilling the contract. All parts and material shall be obtained from the same source(s) of supply as will be used in regular production. Prior to submission, the contractor shall inspect the sample to the degree necessary to ensure that it conforms to the requirements of the contract and submit a record of this inspection with the sample, including statements of findings for materials, processes, and tests. A sample containing known defects to RRS requirements will not be submitted unless specifically authorized by the procuring activity. A first article sample, or portion thereof, as directed by the procuring activity, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days, or whenever a change occurs in manufacturing process, material used, drawing or specification, which significantly affects the product uniformity as determined by the procuring activity.

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4.3.2 Inspections to be performed.- First article test specimens may be subjected by the procuring activity to any or all of the examinations and tests specified in this specification and to any or all requirements of the applicable drawings.

4.3.3 First article rejection.- If any test specimen fails to comply with any of the applicable requirements, the first article samples shall be rejected. The procuring activity reserves the right to terminate its inspection upon failure of any test specimen in the sample to comply with any of the stated requirements.

4.4 Quality conformance inspection.-

4.4.1 Inspection lot.- Unless otherwise specified herein, or by the procuring activity, inspection lot size, lot formation and presentation of lots shall be in accordance with the inspection and sampling procedures herein and the "submission of product" provisions of MIL-STD-105. Lot size shall be no greater than one month's production and RRS assemblies in the lot shall be from one manufacturer and of the same configuration.

4.4.2 Inspections.-

4.4.2.1 Sampling inspection.- Each lot of RRS assemblies shall consist of hardware which has been inspected per Table II. Unless otherwise specified, sampling shall be to Level II of MIL-STD-105. Lots failing to meet the requirements of Table II shall be dispositioned in accordance with the applicable requirements of MIL-STD-105.

4.4.2.2 Lot acceptance testing.- Each lot of RRS assemblies shall be inspected in accordance with Table III. Sample size shall be in accordance with MIL-STD-105, Level S-3, AQL 1.0%. Lots failing to meet these requirements shall be rejected.

4.4.2.2.1 Refurbishment.- RRS assemblies used for lot acceptance testing may be refurbished as required and delivered with the lot they represented. Acceptance inspection of the refurbished RRS assemblies shall be in accordance with Table II.

4.5 Test methods.-

4.5.1 Test conditions.-

4.5.1.1 Environmental conditions.- Unless otherwise specified for the purpose of test, testing shall be conducted at the following conditions:

(a)	Temperature	72° ± 18°F
(b)	Relative Humidity	90 percent maximum
(c)	Barometric Pressure	28 to 32 inches of mercury

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4.5.1.2 Test tolerances.- All tolerances specified herein are absolute with no allowance made for measuring equipment inaccuracy. Measurements shall ensure that the values measured are within the tolerances or limits specified.

4.5.1.3 Inspection equipment.- The inspection equipments required to perform the examinations and tests prescribed herein are described in the "Paragraph Reference/Inspection Method" column in the inspection tables included herein. The contractor shall submit for approval all inspection equipment designs in accordance with the terms of the contract (see 6.4).

4.5.2 Performance tests.-

4.5.2.1 Roll Rate Output Test.- The rate output signal shall be determined for input rates, controlled to ± 5%, of zero, ± 60, ± 120, ± 600, and ± 1800 degrees per second. The output shall meet the requirements of 3.2.1.1.2.1, 3.2.1.1.1.2, and 3.2.1.1.1.3.

4.5.2.2 Roll command output voltage limit test.- The roll command output limit shall be determined by rotating the RRS in the direction corresponding to counterclockwise projectile roll until saturation occurs. The saturated roll command output voltage (negative) shall meet the requirements of 3.2.1.1.2.2. Record the roll attitude and then rotate the RRS in the direction corresponding to clockwise projectile roll until saturation of the opposite polarity occurs. The saturated roll command output voltage (positive), and the roll angle excursion required shall meet the requirements of 3.2.1.1.2.2.

4.5.2.3 Rate signal frequency response test.- The rate signal frequency response shall be measured at the 15 Hz point. The amplitude ratio and the phase shift shall meet the requirements of 3.2.1.2.

4.5.2.4 Null store error due to rate test.- The command signal drift rate or rate signal shall be measured after activation of the RRS at clockwise projectile roll rates, controlled to ± 5%, of zero, 2000, 3000, 3800 and 4350 degrees per second about the input axis. The command signal drift rate at zero roll rate, shall be measured at room temperature, -25°F, and +155°F. The RRS shall meet the requirements specified in 3.2.1.3.

4.5.2.5 Insulation resistance test.- The RRS shall be measured for conformance to the requirements of 3.2.1.4.

4.5.2.6 Activation test.- The RRS shall be spun at a clockwise projectile rate equivalent to a roll output of 3.0 to 4.1 volts and the null store signal shall be removed. The roll command output shall indicate 0.0 ±0.100 volts and shall step up to a saturated level of +9.0 to +14.75 volts within 0.050 seconds after the removal of the null store signal. The roll command output signal shall then step down to 5.7 to 8.5 volts within 0.050 seconds after it stepped up to the initial saturation level, and then ramp upward to the saturation level. The RRS shall be resequenced and spun at a clockwise projectile rate equivalent to a roll output of +1.48 to +2.0 volts

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and the null store signal shall be removed. The roll command output shall be +9.0 to +14.75 volts within 0.050 seconds of removal of null store and shall stay at this value.

4.5.3 Environmental test methods.-

4.5.3.1 High temperature, operating test.- The RRS shall be stabilized at a temperature of +155°F ± 5°F and while maintained at that temperature shall meet the requirements of 3.2.2.1.

4.5.3.2 Low temperature, operating test.- The RRS shall be stabilized at a temperature of -25°F ± 5°F and while maintained at that temperature shall meet the requirements of 3.2.2.1.

4.5.3.3 Vibration test.- The RRS shall be tested in accordance with the vibration environment specified in MIL-STD-810, Method 514.2, Procedure V, Part 2, Table 514.2-V, Figure 514.2-5, Curve AF and meet the requirements of 3.2.2.2.

4.5.3.4 Relative humidity test.- The RRS shall be tested in accordance with the humidity environment specified in MIL-STD-810, Method 507.1, Procedure V, one cycle only, and meet the requirements of 3.2.2.3.

4.5.3.5 Transportation sinusoidal vibration test.- The RRS assembly assembled into a guided projectile, MIL-P-63223, and packaged into a tactical container, MIL-P-63227, shall be subjected to the vibration tests of MIL-STD-810, Method 514.2, Procedure X using the criteria of Table 514.2-VII and Figure 514.2-7, curve AX except search and dwell tests shall be omitted. Exposure of the RRS to this environment shall not be cause for failure of the guidance section in subsequent testing. (See MIL-P-63224).

4.5.3.6 Fragility shock test.- The RRS assembly assembled into a guided projectile, MIL-P-63223, and packaged into a tactical container, MIL-P-63227, shall be subjected to the shock test of MIL-STD-810, Method 516.2, Procedure I, using the shock pulse of Figure 516.2-2 with the exception that the peak acceleration value, 'A', shall be 200 g's and the nominal duration, 'D', shall be 10 milliseconds. Exposure of the RRS to this environment shall not be cause for failure of the guidance section in subsequent testing. (See MIL-P-63224).

4.5.3.7 Launch shock, high temperature test.- The RRS assembly shall be installed in an appropriate canister and the assembly shall be exposed to the launch shock of 3.2.2.6 from an 8 inch gun, M115 or equivalent (GFE), at +145°F ± 2.5°F, (63°C ± 1.4°C) in the forward direction. The launch canister with the RRS assembly installed shall be subjected to the gun launch within 5 minutes after removal from temperature conditioning. After exposure, the RRS assembly shall be examined and shall show no evidence of physical damage and shall meet the performance requirements of 3.2.2.6.

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4.5.3.8 Launch shock, low temperature test.- The RRS assembly shall be installed in an appropriate canister and the assembly exposed to the launch shock of 3.2.2.6 from an 8 inch gun, M115 or equivalent (GFE), at -25°F ± 2.5°F (-32°C ± 1.4°C), in the forward direction. The launch canister with the RRS assembly installed, shall be subjected to the gun launch within 5 minutes after removal from temperature conditioning. After exposure, the RRS assembly shall be examined and shall show no evidence of physical damage and shall meet the performance requirements of 3.2.2.6.

4.5.4 Physical and dimensional characteristics.- The physical and dimensional characteristics shall be inspected to the production drawing requirements of 3.3.1.

4.5.5 Workmanship.- The RRS shall be examined to verify conformance with the workmanship requirements of 3.3.2.

4.5.6 Preparation for delivery inspection.- The preservation, packaging, packing and marking of the RRS for shipping shall be inspected for compliance with the requirements of 5.C.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging and packing.- RRS assemblies shall be preserved, packaged and packed in accordance with the best commercial practices.

5.2 Marking.- The shipping boxes shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use.- The RRS assembly is intended for use in the XM712 Projectile.

6.2 Ordering data.- Procurement documents shall specify the following:

6.2.1 Procurement Requirements.-

(a) Title, number, and date of this specification.

(b) Quantity required and delivery schedules.

(c) Serialization requirements, if applicable.

(d) Quality Conformance Inspection, if other than specified in Section 4 of this specification.

(e) First Article Sample requirements, if other than specified in Section 4 of this specification.

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(f) Packaging requirements, if other than specified in Section 5 of this specification.

(g) Certificates of conformance for each lot or shipment of product.

(h) Requirements for lot control.

6.2.2 Contract data requirements.-

(a) Contract data requirements for inspection equipment designs (conforming to data item description DI-E-1118 or DI-R-1714 with appropriate addendum to each).

(b) Data Cards - When required by incorporation of DD Form 1423 into the contract, data cards shall be prepared for each lot in accordance with Data Item description DI-L-1410, except that for propellant or explosive lots, Acceptance and Description Sheets will be prepared in accordance with MIL-STD-1171.

6.3 Definitions.-

6.3.1 Contractor.- Unless otherwise specified, the word contractor used throughout this specification shall mean any supplier or producer of items or material to the Government. This shall include commercial contractors, sub-contractors, Government Owned, Contractor Operated (GOCO) plants.

6.3.2 Technical Data Package (TDP).- A technical description of an item adequate for use in procurement. The description defines the required design configuration and assures adequacy of item performance. It consists of all applicable technical data such as plans, drawings, and associated lists of specifications, standards, models, performance requirements, quality assurance provisions, and packaging data.

6.3.3 Inspection.- The contractor will not assume, nor does the Government guarantee that all possible combinations permitted by the tolerance limits of the specifications and drawings will consistently satisfy the test requirements. Therefore, the manufacturer is obligated to choose those combinations of tolerance and fit within the limits of the specification and drawings that best suit his process needs and still satisfy the testing requirements.

6.4 Inspection equipment designs.- Inspection equipment designs are of two types - Government Special Inspection Equipment (SIE) designs and contractor designs. SIE designs are designated by drawing numbers under the "Method of Inspection" heading in Tables I, II, and III, or on the equipment lists referenced on the equipment tabulation, whichever is specified in the item detail specification. Design responsibility for all other inspection equipment is assigned to the contractor. The contractor need not furnish any design when a complete Government SIE design is provided. Unless

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otherwise specified, however, the contractor may submit alternate or modified contractor designs of SIE in accordance with 6.4.2 and 6.4.3 should he elect to do so.

6.4.1 SIE Designs.- SIE designs may consist of any of the following:

(a) Detailed drawings which completely depict all information necessary for the fabrication and use of the item of inspection equipment.

(b) A source control drawing or a specification control drawing as defined in MIL-STD-100.

(c) An envelope drawing, as defined in MIL-STD-100, which establishes the criteria which a detailed design shall meet. When the envelope drawings are specified, the contractor shall prepare designs which comply with the criteria therein.

6.4.2 Contractor designs.- Contractor designs are required for all inspection equipment for which SIE designs are not specified and may include commercial equipment which the contractor proposes to use. (Commercial equipment is defined as unmodified equipment which is cataloged and available for purchase by the general public). Contractor designs shall include appropriate operating instructions, calibration procedures and maintenance procedures. Commercial equipment shall be fully described by catalog listings or other means which provide sufficient information to permit identification and evaluation by the Government and may include illustrations and engineering data. Designs shall be prepared for any special fixture(s) required to be used with commercial equipment, or with SIE designs if not otherwise covered.

6.4.3 Submission of designs for approval.- Contractor designs shall be approved by the Government prior to fabricating or procuring the equipment. Designs shall be submitted for approval in accordance with the stipulations, time frame and distribution specified in the Contract Data Requirements Lists (DD Form 1423). Partial submission of inspection equipment designs is permissible and encouraged. However, the completion date for design review will be based on the date of the final submission of designs and the required delivery schedule as stipulated in the contract. The specified segment of Picatinny Arsenal to which the contractor designs shall be sent will be specified in the item detail specification.

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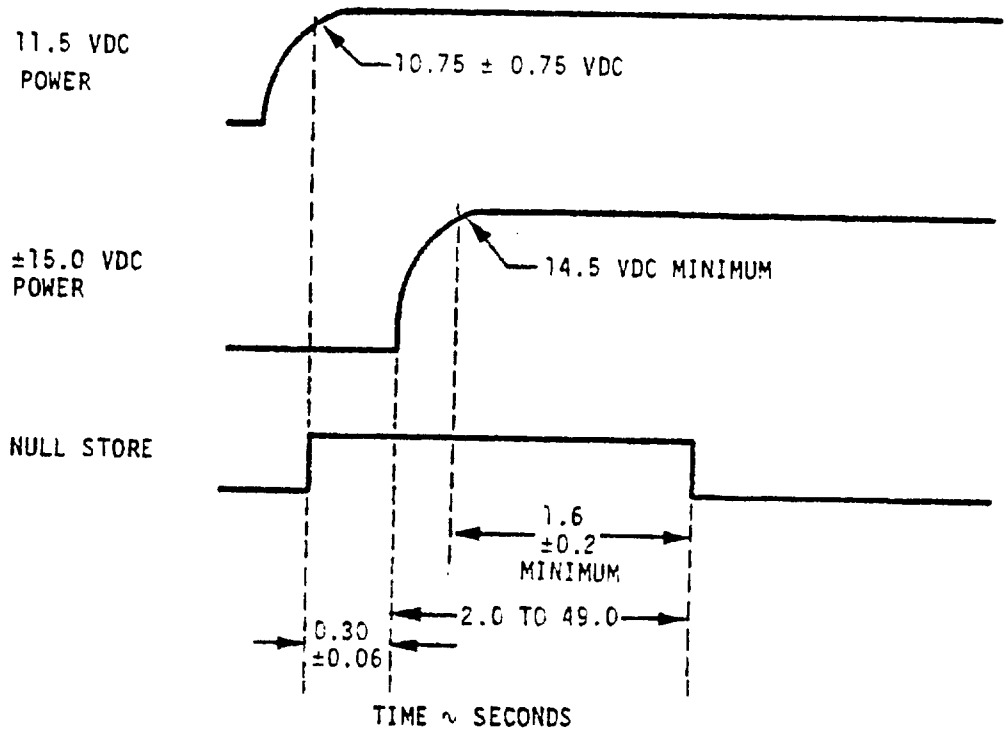


Figure 1. Timing Sequence

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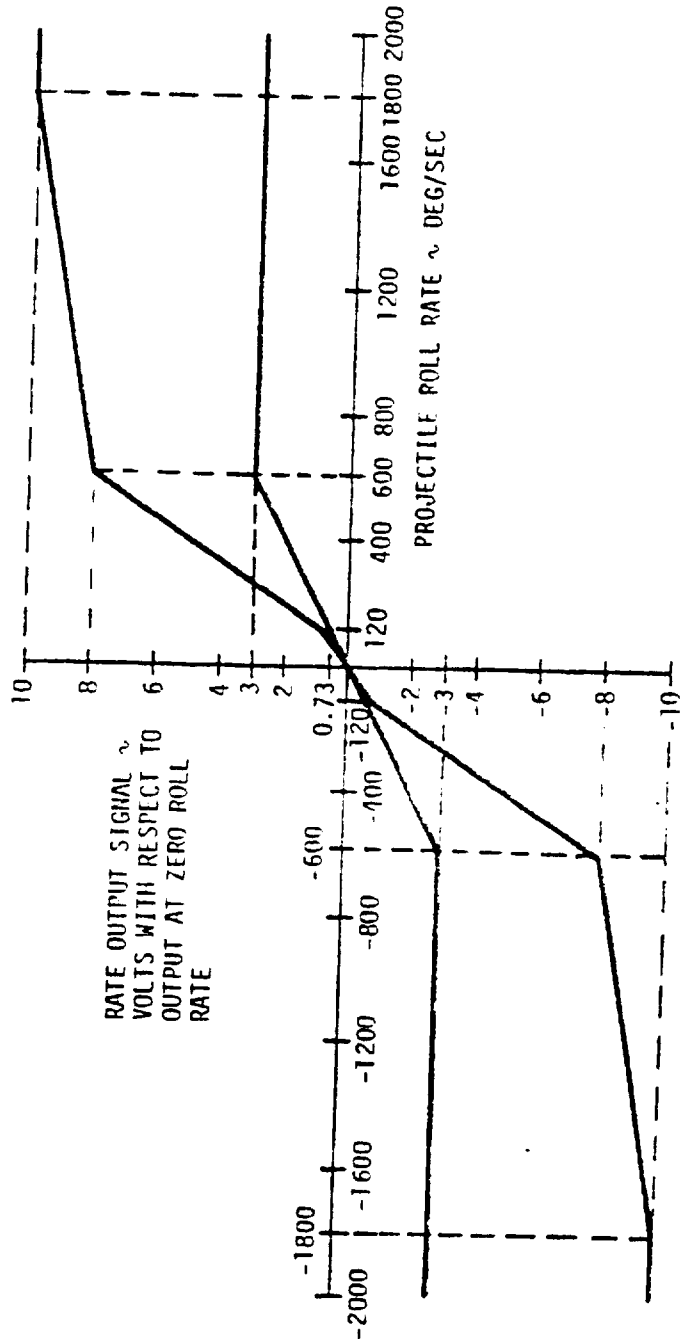


Figure 2. Roll Rate Output Boundaries

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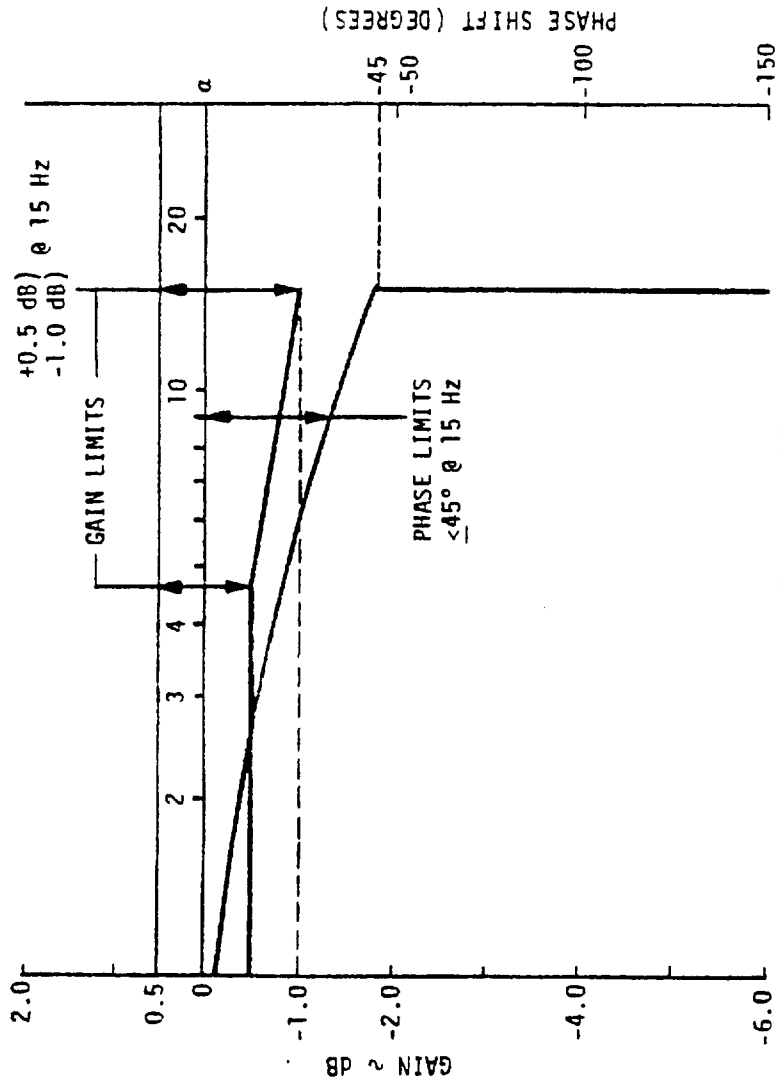


Figure 3. Rate Output Amplitude and Phase Limits

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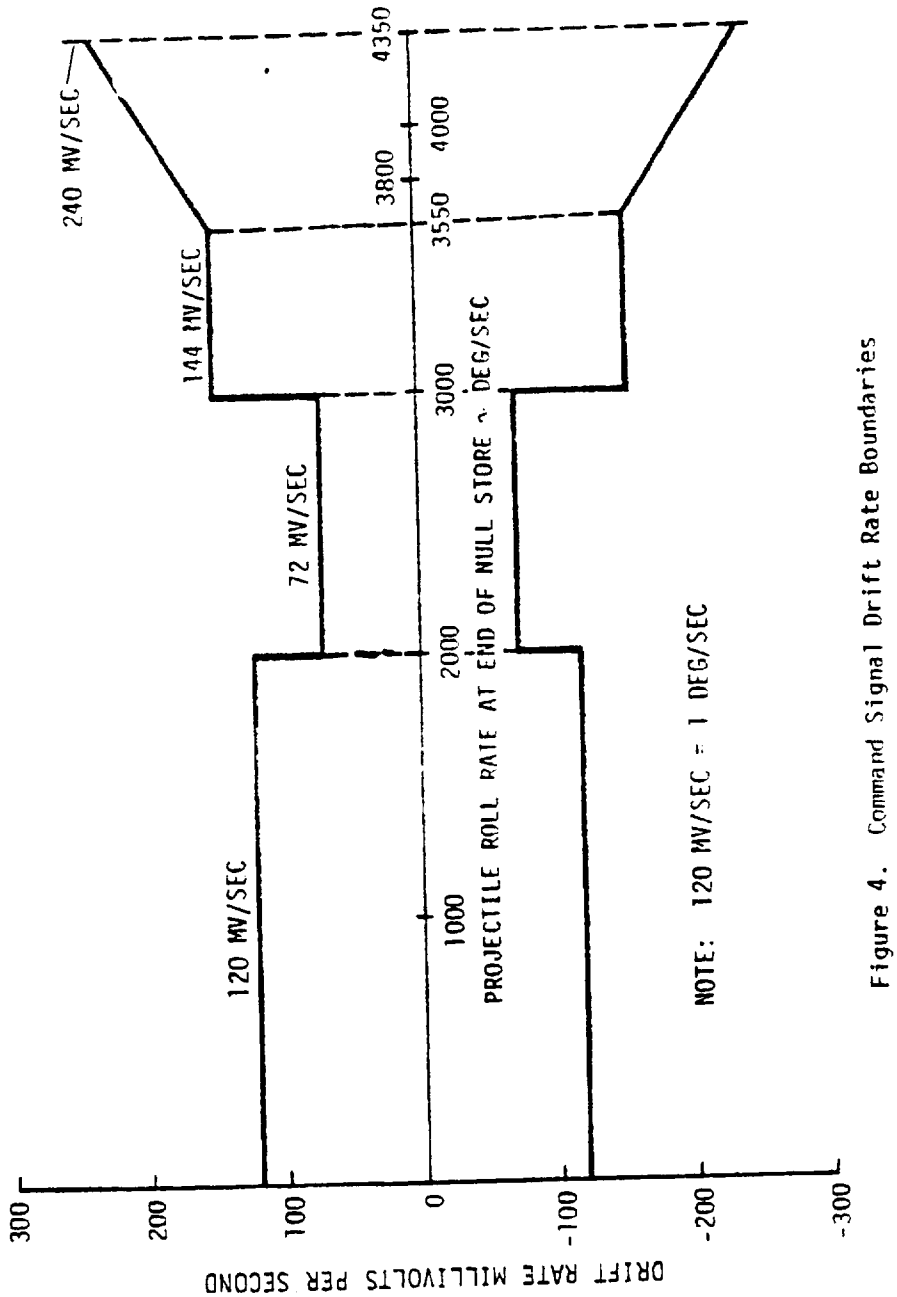


Figure 4. Command Signal Drift Rate Boundaries

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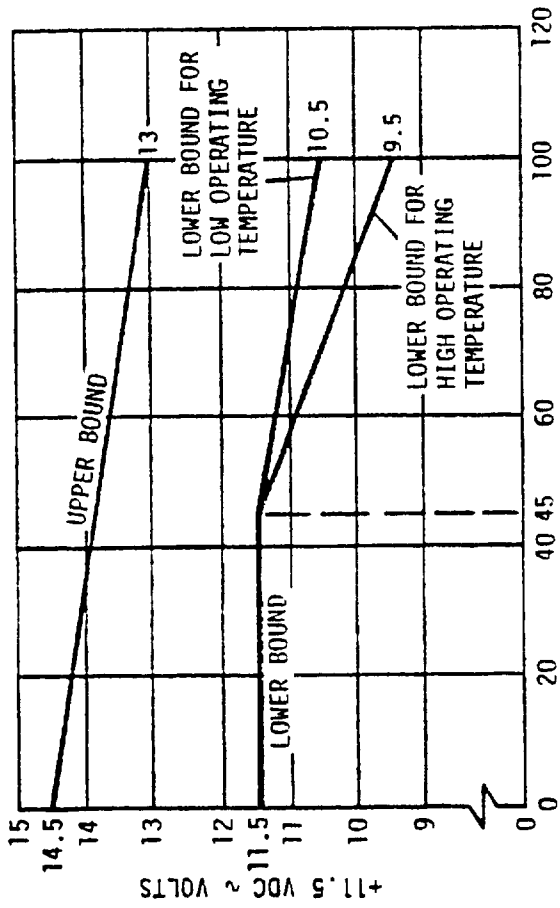


Figure 5. Time Varying Characteristics of 11.5 Volt Input Power

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TABLE I
FIRST ARTICLE INSPECTION
CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 1		DRAWING NUMBER
	ROLL RATE SENSOR ASSEMBLY (See Note 2)			9303350
				NEXT HIGHER ASSEMBLY
				9306574
CATEGORY	EXAMINATION OR TEST	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE INSPECTION METHOD
Critical	None defined			
Major				
101	Performance (Note 1)	100%	3.2.1	4.5.2
102	Vibration	100%	3.2.2.2	4.5.3.3
103	Relative Humidity	100%	3.2.2.3	4.5.3.4
104	Launch Shock (Note 1)	100%	3.2.2.6	4.5.3.7, 4.5.3.8
105	Performance at Ambient Temperature	100%	3.2.1	4.5.2
106	Transportation Sinusoidal Vibration	100%	3.2.2.4	4.5.3.5
107	Fragility Shock	100%	3.2.2.5	4.5.3.6
Minor	None defined			

NOTES: (1) Three at 155°F and three at -25°F.
(2) Major items 101 through 105 shall be run on 6 units only; major items 106 and 107 shall be conducted on 10 additional units as part of higher level procuring activity testing.

MIL-P-83234(AR)

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TABLE II
CONFORMANCE INSPECTIONS, SAMPLING
CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE: BOARD ASSEMBLY, PUMP OSCILLATOR	SHEET 1 OF 7		DRAWING NUMBER: 9303368
				NEXT HIGHER ASSEMBLY: 9303378
CATEGORY	EXAMINATION OR TEST	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE INSPECTION METHOD
Critical	None defined			
Major				
101	Temperature Shock	100%	Dwg	Dwg
102	Acceptance Test	100%	Dwg	9303370
Minor				
201	Workmanship	0.65%	3.3.2	4.5.5

NOTES:

MIL-P-63234(AR)

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TABLE II
CONFORMANCE INSPECTIONS, SAMPLING
CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE: BOARD ASSEMBLY, BRIDGE AMPLIFIER	SHEET 2 OF 7		DRAWING NUMBER: 9303366
				NEXT HIGHER ASSEMBLY: 9303378
CATEGORY	EXAMINATION OR TEST	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE INSPECTION METHOD
Critical	None defined			
Major				
101	Temperature Shock	100%	Dwg	Dwg.
102	Acceptance Test	100%	Dwg	9303371
Minor				
201	Workmanship	0.65%	3.3.2	4.5.5

NOTES:

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TABLE II
CONFORMANCE INSPECTIONS, SAMPLING
CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE: PROCESSOR ASSEMBLY	SHEET 3 OF 7		DRAWING NUMBER: 9303364
				NEXT HIGHER ASSEMBLY: 9303378
CATEGORY	EXAMINATION OR TEST	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE INSPECTION METHOD
Critical	None defined			
Major				
101	Temperature Shock	100%	Dwg	Dwg
102	Acceptance Test	100%	Dwg	9303372
Minor				
201	Workmanship	0.65%	3.3.2	4.5.5

NOTES:

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TABLE II
CONFORMANCE INSPECTIONS, SAMPLING
CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 4 OF 7		DRAWING NUMBER
	HOUSING, RATE SENSOR			9303382
				NEXT HIGHER ASSEMBLY
				9303350
CATEGORY	EXAMINATION OR TEST	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE INSPECTION METHOD
Critical	None defined			
Major				
101	Outline dimensions	0.40%	9303390	Gage
102	Mounting Hole Locations	0.40%	9303390	Gage
Minor				
201	Workmanship	0.65%	3.3.2	4.5.5

NOTES:

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TABLE II
CONFORMANCE INSPECTIONS, SAMPLING
CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 5 OF 7		DRAWING NUMBER
	COVER, RATE SENSOR ASSEMBLY			9303362
CATEGORY	EXAMINATION OR TEST	AQL OR 100%	REQUIREMENT PARAGRAPH	NEXT HIGHER ASSEMBLY 9303350 / PARAGRAPH REFERENCE INSPECTION METHOD
Critical	None defined			
Major				
101	Outline dimensions	0.40%	Dwg	Gage
102	Thickness	0.40%	Dwg	Gage
Minor				
201	Workmanship	0.65%	3.3.2	4.5.5

NOTES:

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CONFORMANCE INSPECTIONS, SAMPLING
CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE: BLOCK, MOUNTING	SHEET 6 OF 7		DRAWING NUMBER: 9303380
				NEXT HIGHER ASSEMBLY: 9303350
CATEGORY	EXAMINATION OR TEST	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE INSPECTION METHOD
Critical	None defined			
Major	None defined			
Minor				
201	Mounting hole location	0.65%	Dwg	Gage
202	Internal diameter	0.65%	Dwg	Gage
203	Height of step at base	0.65%	Dwg	Gage
204	Width of step at base	0.65%	Dwg	Gage
205	Workmanship	0.65%	3.3.2	4.5.5

NOTES:

MIL-P-03534(AR)

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TABLE II
CONFORMANCE INSPECTIONS, SAMPLING
CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 7 OF 7		DRAWING NUMBER
	ROLL RATE SENSOR ASSEMBLY			9303350
				NEXT HIGHER ASSEMBLY
				9306574
CATEGORY	EXAMINATION OR TEST	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE INSPECTION METHOD
Critical	None defined			
Major				
101	Roll rate output	100%	3.2.1.1.2.1	4.5.2.1
102	Roll command output	100%	3.2.1.1.2.2	4.5.2.2
103	Rate signal frequency response	100%	3.2.1.2	4.5.2.3
104	Combined signal drift rate	100%	3.2.1.3	4.5.2.4
105	Insulation resistance	100%	3.2.1.4	4.5.2.5
Minor				
201	Workmanship	0.65%	3.3.2	4.5.5
202	Preparation for delivery	0.65%	5.0	4.5.6

NOTES:

MIL-P-63234(AR)

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TABLE III
LOT ACCEPTANCE TESTS
CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 1		DRAWING NUMBER
	ROLL RATE SENSOR ASSEMBLY			9303350
				NEXT HIGHER ASSEMBLY
				9306574
CATEGORY	EXAMINATION OR TEST	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE INSPECTION METHOD
Critical	None defined			
Major				
101	Vibration	1.0%	3.2.2.2	4.5.3.3
102	Performance (Note 1)	1.0%	3.2.1	4.5.2
Minor	None defined			

NOTES:
(1) One half or next lowest integer shall be tested at 155°F and remainder at -25°F.

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-P-63234 Projectile, Guided, 155MM, XM712 Roll Rate Sensor For

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

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